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Section-6

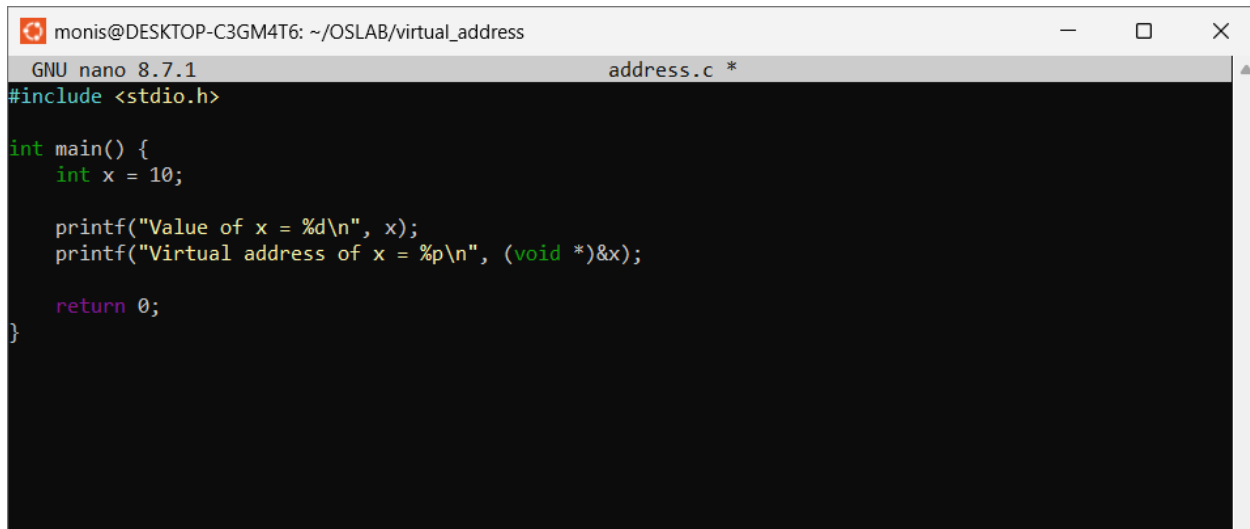
Practical Programs

1.Virtual Address

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
monis@DESKTOP-C3GM4T6:~$ ls
nux-System-Programming-Week1  OSSP-Practical      foldername  moni      sample1.c
LAB                            Simple-Commands-Week1 hello       monis     sample1.c.save
LAB14                          example.txt          hello.c     sample1
monis@DESKTOP-C3GM4T6:~$ cd OSLAB
monis@DESKTOP-C3GM4T6:~/OSLAB$ mkdir virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB$ ls
LAB28          file1.txt  hello.c:Zone.Identifier  hello_shortcut  test1
pes            hello      hello:Zone.Identifier    newfile.txt    virtual_address
heduling-Algorithms hello.c    hello_link               skill
monis@DESKTOP-C3GM4T6:~/OSLAB$ cd virtual_address
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ nano address.c
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$
```

1. Complete code

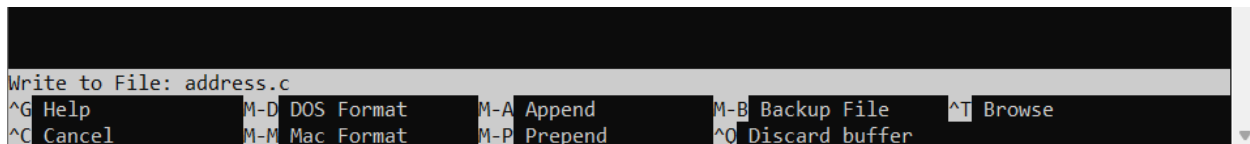


```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
GNU nano 8.7.1 address.c *
#include <stdio.h>

int main() {
    int x = 10;

    printf("Value of x = %d\n", x);
    printf("Virtual address of x = %p\n", (void *)&x);

    return 0;
}
```

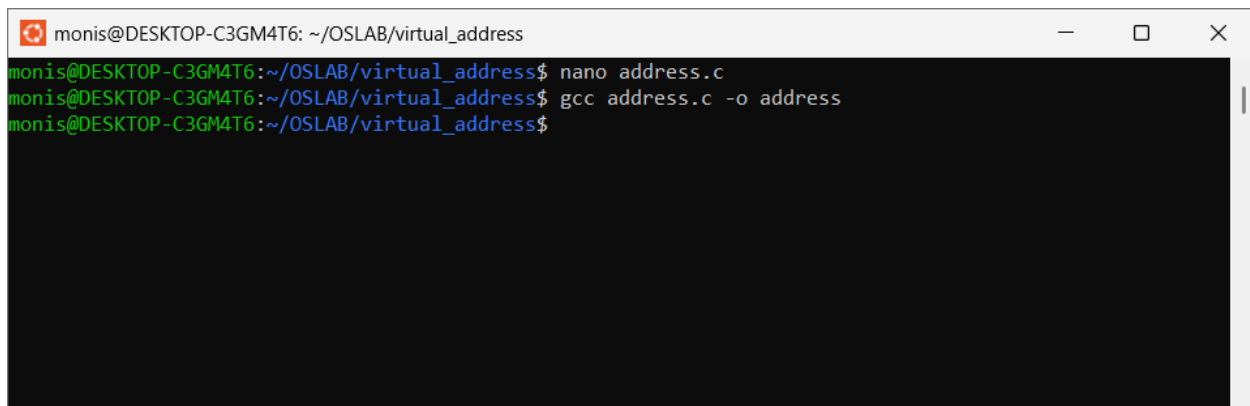


```
Write to File: address.c
^G Help      M-D DOS Format  M-A Append     M-B Backup File ^T Browse
^C Cancel    M-M Mac Format  M-P Prepend    ^O Discard buffer
```

3. Compile the program

If you are using **GCC**, open Command Prompt/Terminal in the folder containing address.c and run:

```
gcc address.c -o address
```



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ nano address.c
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ gcc address.c -o address
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$
```

4. Run the program

On Windows:

address

or:

.\address.exe

On Linux/macOS:

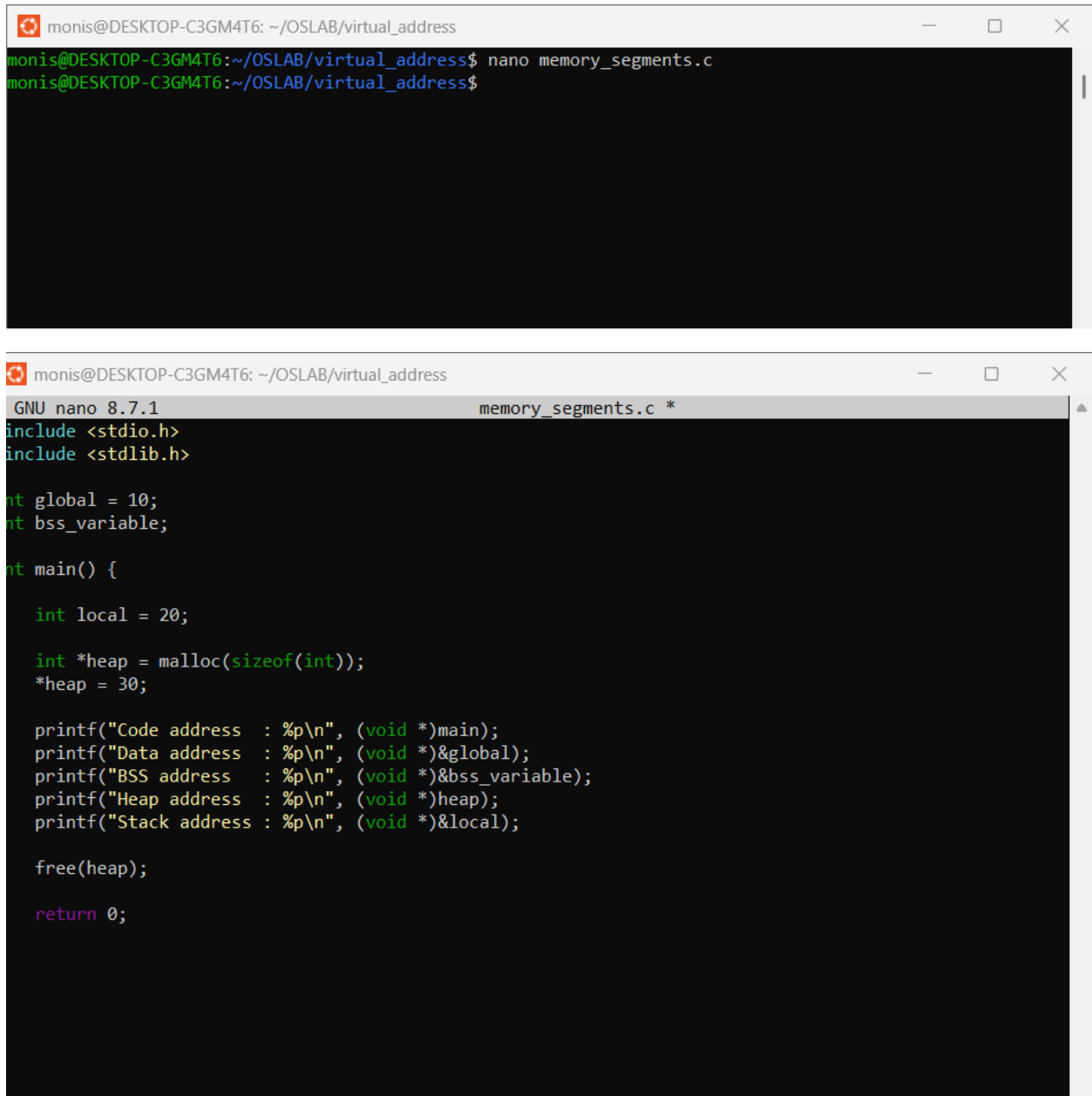
./address

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ nano address.c
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ gcc address.c -o address
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ address
address: command not found
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ .\address.exe
.address.exe: command not found
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ ./address
Value of x = 10
Virtual address of x = 0x7ffe2021ba44
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ .\address.exe
.address.exe: command not found
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ ./address
Value of x = 10
Virtual address of x = 0x7ffd04ec9864
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$
```

2.Memory Segments

Step 1 — Create the file



The first screenshot shows a terminal window with the prompt `monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address`. The user has entered the command `nano memory_segments.c`, and the prompt is now `monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$`.

The second screenshot shows the same terminal window with the `nano` editor open. The title bar indicates the file is `memory_segments.c *`. The editor shows the following code:

```
GNU nano 8.7.1 memory_segments.c *
#include <stdio.h>
#include <stdlib.h>

int global = 10;
int bss_variable;

int main() {

    int local = 20;

    int *heap = malloc(sizeof(int));
    *heap = 30;

    printf("Code address : %p\n", (void *)main);
    printf("Data address : %p\n", (void *)&global);
    printf("BSS address : %p\n", (void *)&bss_variable);
    printf("Heap address : %p\n", (void *)heap);
    printf("Stack address : %p\n", (void *)&local);

    free(heap);

    return 0;
}
```

Step 2 — Save in nano

Press:

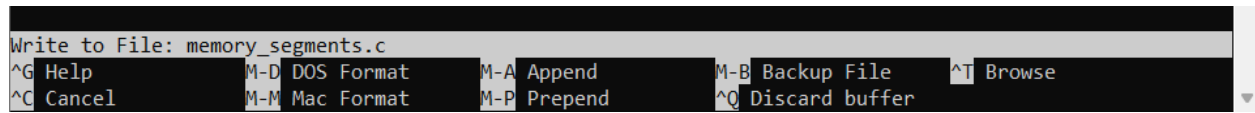
Ctrl + O

Then press:

Enter

Then:

Ctrl + X

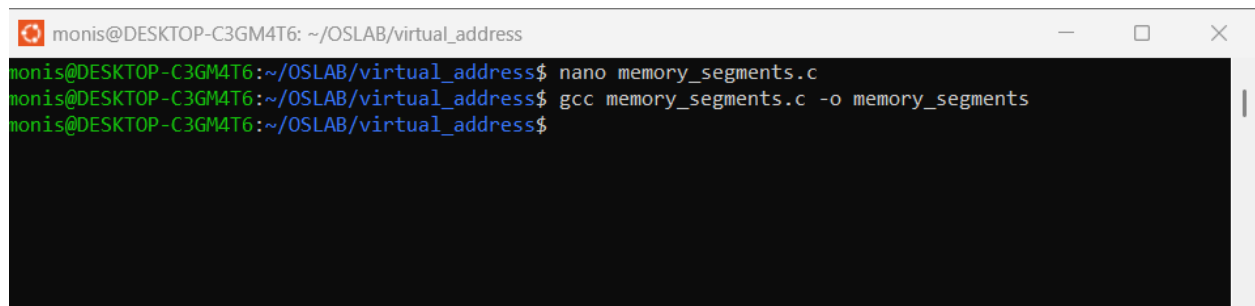


Step 3 — Compile

Run:

gcc memory_segments.c -o memory_segments

If there is no error, compilation is successful.



Step 4 — Run

Because you're using Ubuntu/WSL, use:

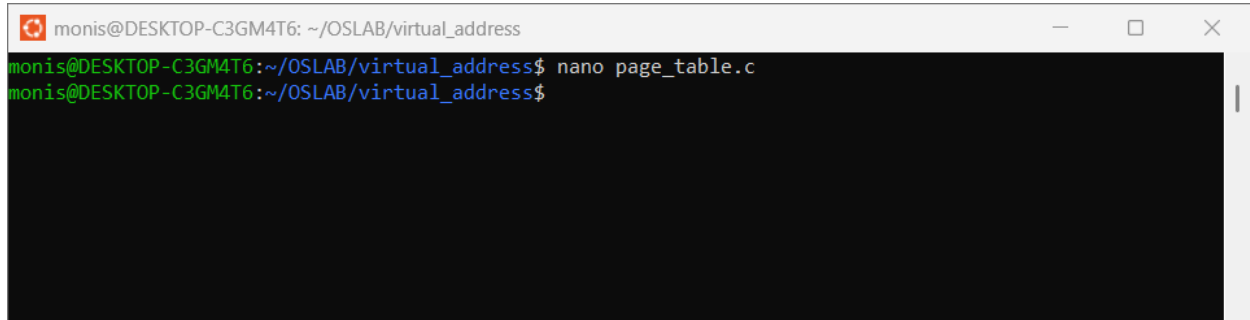
./memory_segments

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ nano memory_segments.c
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ gcc memory_segments.c -o memory_segments
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ ./memory_segments

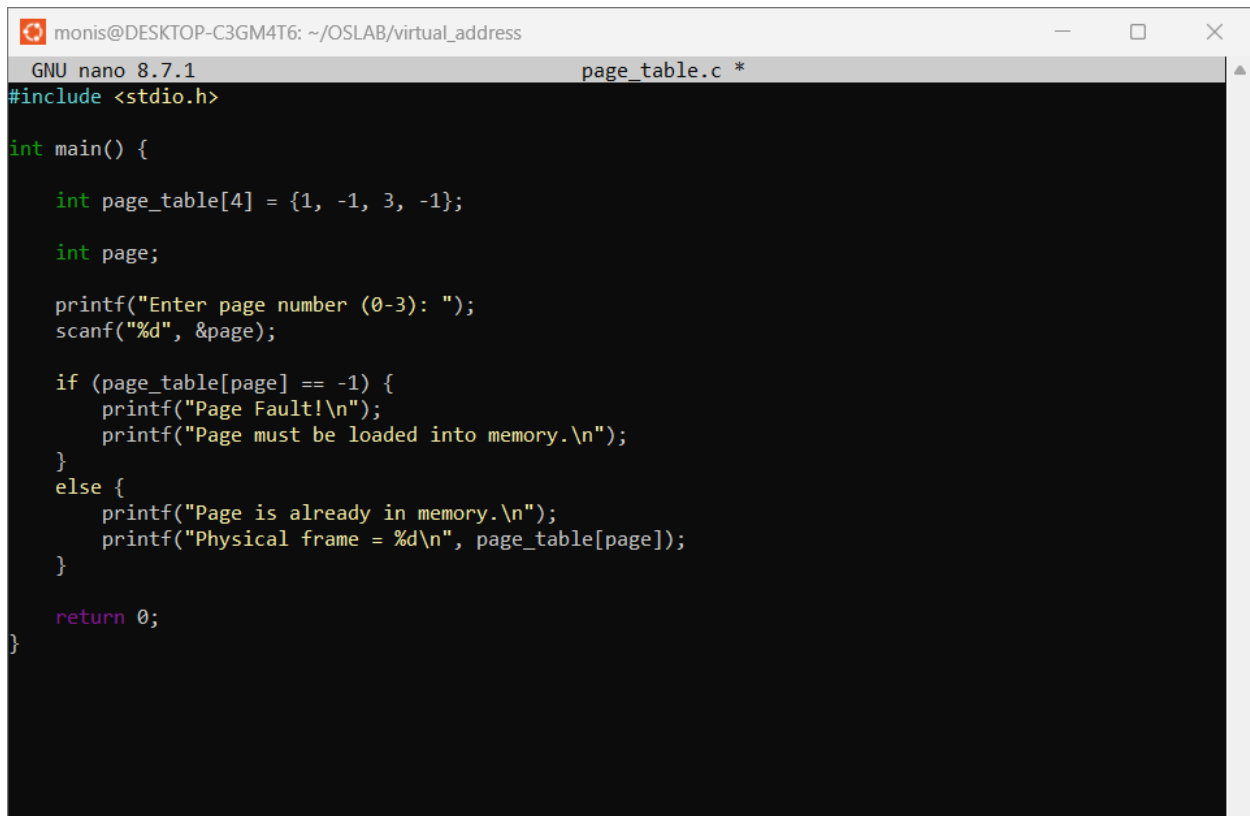
Code address : 0x5a745b3591a9
Data address : 0x5a745b35c010
BSS address  : 0x5a745b35c018
Heap address : 0x5a745fca7010
Stack address: 0x7ffff64bfe7c
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$
```

3. Page Table

1. Create the C file



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ nano page_table.c
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$
```



```
GNU nano 8.7.1 page_table.c *
#include <stdio.h>

int main() {

    int page_table[4] = {1, -1, 3, -1};

    int page;

    printf("Enter page number (0-3): ");
    scanf("%d", &page);

    if (page_table[page] == -1) {
        printf("Page Fault!\n");
        printf("Page must be loaded into memory.\n");
    }
    else {
        printf("Page is already in memory.\n");
        printf("Physical frame = %d\n", page_table[page]);
    }

    return 0;
}
```

2. Save and exit nano

Press:

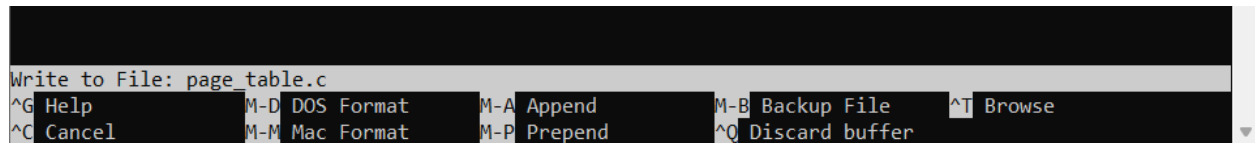
Ctrl + O

Then:

Enter

Then:

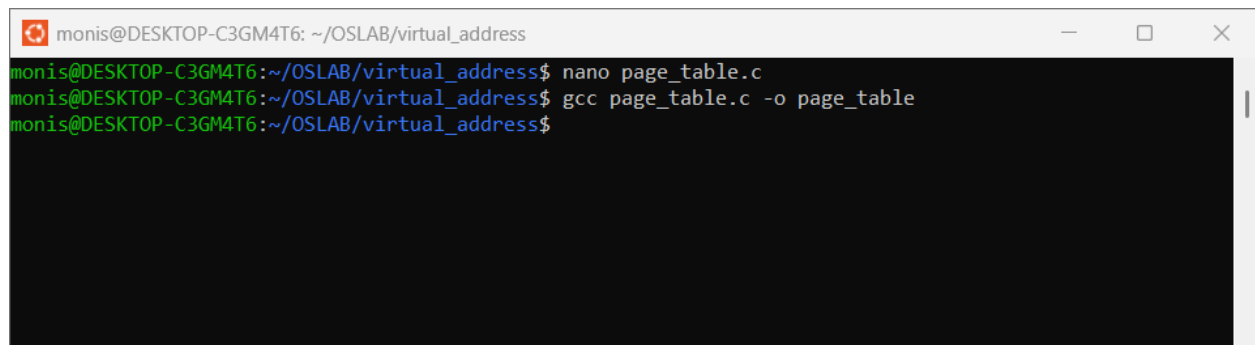
Ctrl + X



3. Compile

Run:

gcc page_table.c -o page_table



4. Run

./page_table

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ nano page_table.c
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ gcc page_table.c -o page_table
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ ./page_table
Enter page number (0-3): 0
Page is already in memory.
Physical frame = 1
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$
```

5. Test page 1

Run again:

`./page_table`

Enter:

1

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ ./page_table
Enter page number (0-3): 0
Page is already in memory.
Physical frame = 1
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ ./page_table
Enter page number (0-3): 1
Page Fault!
Page must be loaded into memory.
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$
```

6. Test all four pages

Input 0:

Page is already in memory.

Physical frame = 1

Input 1:

Page Fault!

Page must be loaded into memory.

Input 2:

Page is already in memory.

Physical frame = 3

Input 3:

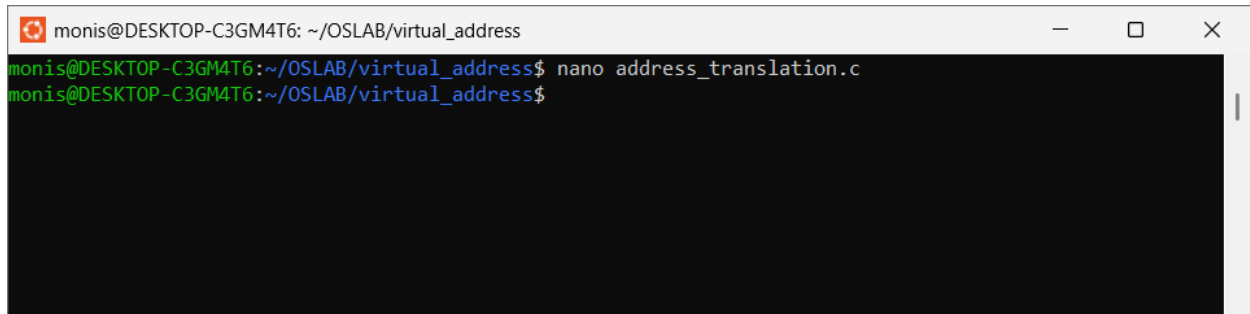
Page Fault!

Page must be loaded into memory.

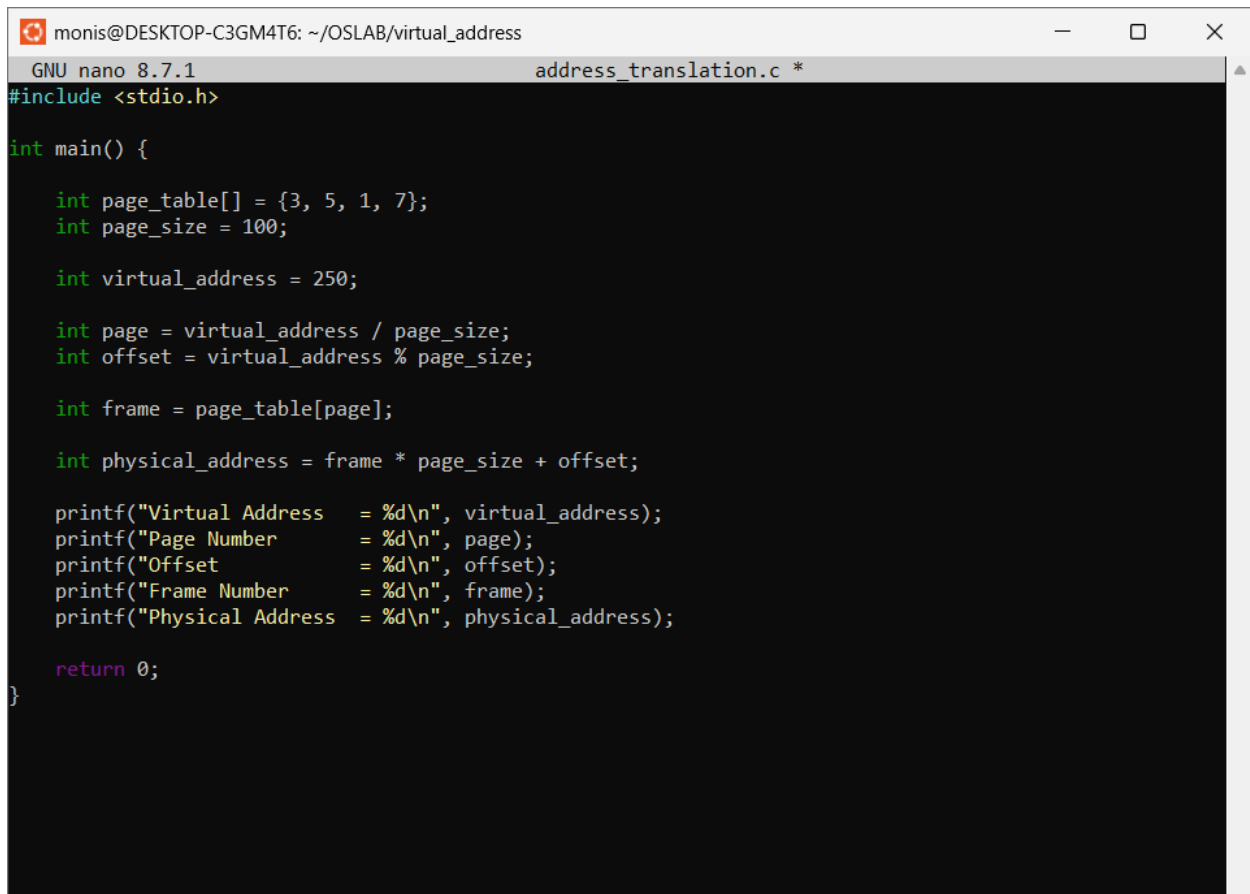
```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ ./page_table
Enter page number (0-3): 0
Page is already in memory.
Physical frame = 1
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ ./page_table
Enter page number (0-3): 1
Page Fault!
Page must be loaded into memory.
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ ./page_table
Enter page number (0-3): 2
Page is already in memory.
Physical frame = 3
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ ./page_table
Enter page number (0-3): 3
Page Fault!
Page must be loaded into memory.
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$
```

4.Address Translation

1. Create the file



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ nano address_translation.c
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$
```



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
GNU nano 8.7.1 address_translation.c *
#include <stdio.h>

int main() {

    int page_table[] = {3, 5, 1, 7};
    int page_size = 100;

    int virtual_address = 250;

    int page = virtual_address / page_size;
    int offset = virtual_address % page_size;

    int frame = page_table[page];

    int physical_address = frame * page_size + offset;

    printf("Virtual Address    = %d\n", virtual_address);
    printf("Page Number       = %d\n", page);
    printf("Offset            = %d\n", offset);
    printf("Frame Number      = %d\n", frame);
    printf("Physical Address   = %d\n", physical_address);

    return 0;
}
```

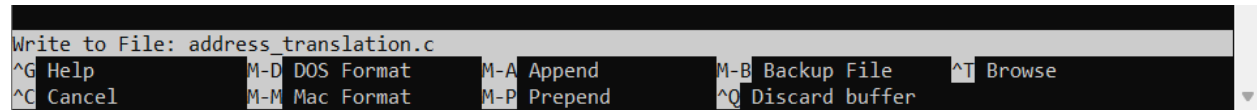
2. Save and exit

In nano:

Ctrl + O

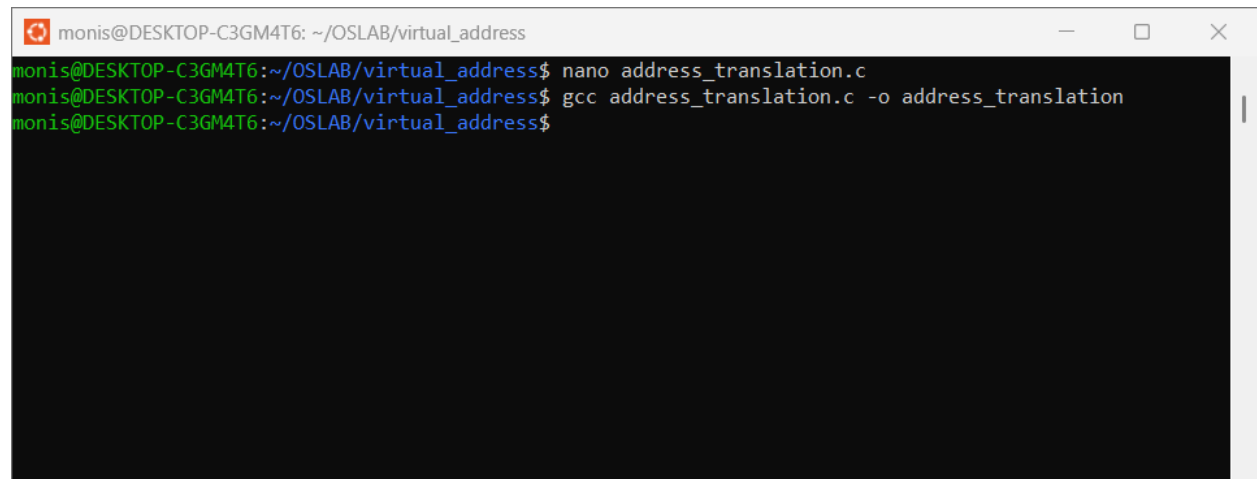
Press Enter, then:

Ctrl + X



3. Compile

gcc address_translation.c -o address_translation



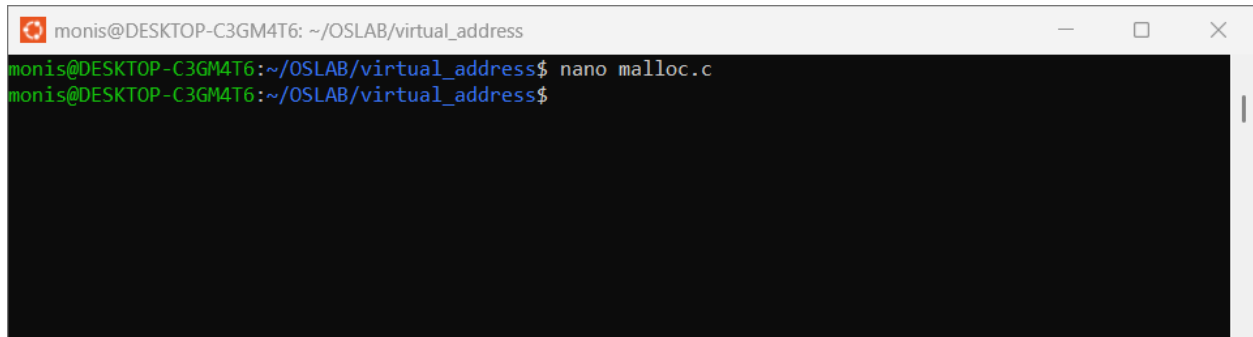
4. Run

./address_translation

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ nano address_translation.c
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ gcc address_translation.c -o address_translation
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ ./address_translation
Virtual Address = 250
Page Number    = 2
Offset         = 50
Frame Number   = 1
Physical Address = 150
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$
```

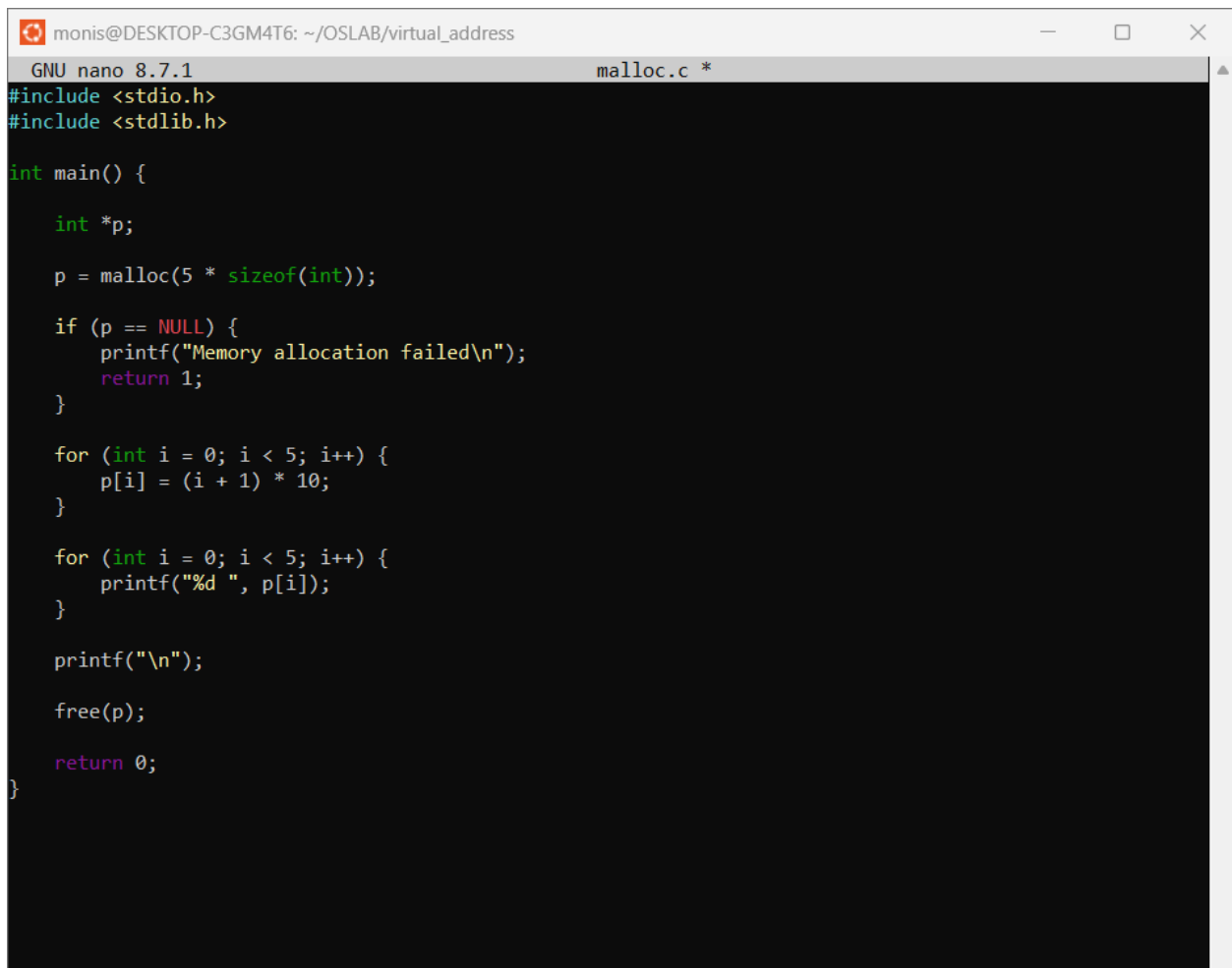
5.Dynamic Memory(malloc)

1. Create the file



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ nano malloc.c
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$
```

2. Paste the malloc program



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
GNU nano 8.7.1 malloc.c *
#include <stdio.h>
#include <stdlib.h>

int main() {
    int *p;

    p = malloc(5 * sizeof(int));

    if (p == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }

    for (int i = 0; i < 5; i++) {
        p[i] = (i + 1) * 10;
    }

    for (int i = 0; i < 5; i++) {
        printf("%d ", p[i]);
    }

    printf("\n");

    free(p);

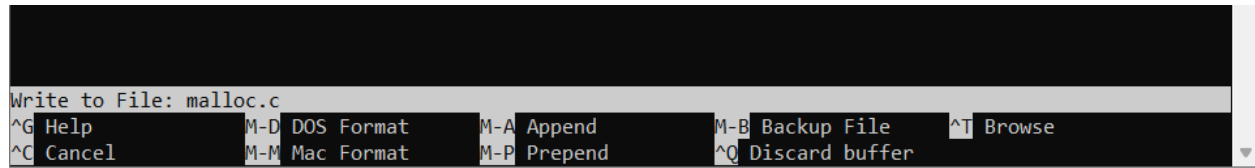
    return 0;
}
```

3. Save

Ctrl + O

Enter

Ctrl + X



4. Compile

gcc malloc.c -o malloc

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ nano malloc.c
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ gcc malloc.c -o malloc
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$
```

5. Run

./malloc

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ nano malloc.c
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ gcc malloc.c -o malloc
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ ./malloc
0 20 30 40 50
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$
```

6. Segmentation Fault

1. Create the file

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ nano segmentation_fault.c
```

2. Paste the program

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
GNU nano 8.7.1 segmentation_fault.c *
#include <stdio.h>

int main() {
    int *p = NULL;

    printf("Before accessing memory\n");

    *p = 100;

    printf("After accessing memory\n");

    return 0;
}
```

3. Save

Ctrl + O

Enter

Ctrl + X

```
Write to File: segmentation_fault.c
^G Help      M-D DOS Format  M-A Append     M-B Backup File ^T Browse
^C Cancel    M-M Mac Format  M-P Prepend    ^O Discard buffer
```

4. Compile:

gcc segmentation_fault.c -o segmentation_fault

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ nano segmentation_fault.c
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ gcc segmentation_fault.c -o segmentation_fault
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$
```

5.Run:

./segmentation_fault

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ nano segmentation_fault.c
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ gcc segmentation_fault.c -o segmentation_fault
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$ ./segmentation_fault
Before accessing memory
Segmentation fault      ./segmentation_fault
monis@DESKTOP-C3GM4T6:~/OSLAB/virtual_address$
```



